

Final: Full Approach

Team Name: Deep Learners

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During the Final project, Deep Learners adopted different approaches to achieve the desired results. These approaches are listed with details as follows:

Model/Evaluation Approach:

For the final project, which is an extension of the Deep Learners Mid-Term, their task was to build, evaluate, and compare multiple Machine Learning (ML) models to predict the number of purchases made in the last month. We applied a consistent modeling and evaluation pipeline to ensure fair comparison across the five models.

Model Data Prep:

The completed dataset included categorical, numerical, and engineered features derived from product pricing, reviews, discounts, ratings, and category information. Our target variable is defined as `purchased_last_month`. Administrative columns and non-predictive text were removed. Also, rows with missing target values were excluded to ensure valid model metric evaluation and training.

Train/Validation/Test Split:

We divided the dataset into three subsets to support comparisons of object performance. The split was 70 percent training, 15 percent validation, and 15 percent test. We used the training set to fit the models, the validation set to compare and select models, and the test set to confirm final performance.

Preprocessing Pipeline:

Our preprocessing pipeline was unified and applied to all models. Numerical and scaled features were imputed using the median and standard normalization. Before being

converted into numerical form using one-hot encoding, categorical features were imputed with the most frequent values. To avoid data leakage, the preprocessing pipeline was fit only on the training data and then applied to the validation and test sets.

Model Training:

Our five regression models, Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and K Nearest Neighbor (KNN) Regressor, were all trained using the preprocessed training data.

Model Evaluation Metrics:

The model's performance was evaluated using standard regression metrics, including Mean Absolute Error (MAE), which measures the average prediction error; Mean Squared Error (MSE), which penalizes large errors; and R^2 (Coefficient of Determination), which measures the proportion of variance explained. The metrics were computed for both the validation and test sets using the predictions generated by each model.

Model Comparison:

Summary tables clearly display MAE, MSE, and R^2 values for all models. Comparing allowed us to identify models with lower prediction error and higher explanatory power. To support interpretation, visualizations were used, including comparison charts and plots of predicted versus actual values.

Ensemble Modeling:

The deep Learners implemented ensemble approaches to evaluate whether combining multiple models improved performance, including simple averaging of top-performing models and voting-based ensembles. We evaluated the ensemble models using the same metrics as the base models to ensure consistency through our framework.

Conclusion:

Our final comparisons demonstrated apparent performance differences across modeling strategies and models. The models were selected and ranked based on validation performance, using the test set to confirm generalization. Taking this approach, the Deep Learners ensure that conclusions are data-driven, reproducible, and grounded in standard machine learning evaluation practices.